Algorithm Sample 1

Dataset-expansion sample

Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

1. For $\beta \in \left(1, \frac{3}{2}\right)$ set $\varepsilon = \frac{3}{2} - \beta$ and $T = L^{2(1-\varepsilon)}$. For $i = 1, ..., d$, solve for the approximate first-order corrector $\varphi_{i,T}^{(L)}$:

$$\frac{1}{T}\varphi_{i,T}^{(L)} - \nabla \cdot a\nabla\varphi_{i,T}^{(L)} = \nabla \cdot ae_i \quad \text{in} \;\; Q_{2L}, \qquad \varphi_{i,T}^{(L)} = 0 \quad \text{on} \;\; \partial Q_{2L}.$$

2. Calculate the approximate homogenized coefficients via

$$a_h^{(L)} e_i = \int \omega q_{i,T}^{(L)},$$

where

$$q_{i,T}^{(L)} := a\Big(e_i + \nabla\varphi_{i,T}^{(L)}\Big)$$

and $\omega(x) = \frac{1}{L^d}\hat{\omega}\left(\frac{x}{L}\right)$ with $\hat{\omega}$ as in Theorem 1.

3. Find $\tilde{u}_h^{(L)}$ on ∂Q_L:

$$\tilde{u}_h^{(L)} = \int G_h^{(L)} * (\nabla \cdot g),$$

where $G_h^{(L)}(x) := \frac{1}{4\pi \left|\left(a_h^{(L)}\right)^{-1/2} x\right|}$ is the Green function for the constant-coefficient operator $-\nabla \cdot a_h^{(L)} \nabla$.

4. Solve for approximate first-order flux correctors $\sigma_{i,T}^{(L)} = \left\{\sigma_{ijk,T}^{(L)}\right\}_{j,k}$:

$$\frac{1}{T}\sigma_{ijk,T}^{(L)} - \Delta\sigma_{ijk,T}^{(L)} = \partial_j q_{ik,T}^{(L)} - \partial_k q_{ij,T}^{(L)} \quad \text{in} \;\; Q_{\frac{7}{4}L}, \qquad \sigma_{ijk,T}^{(L)} = 0 \quad \text{on} \;\; \partial Q_{\frac{7}{4}L}.$$

5. Solve for approximate second-order correctors $\psi_{ij,T}^{(L)}$:

$$\frac{1}{T}\psi_{ij,T}^{(L)} - \nabla \cdot a\nabla\psi_{ij,T}^{(L)} = \nabla \cdot \Big(\varphi_{i,T}^{(L)} a - \sigma_{i,T}^{(L)}\Big)e_j \quad \text{in} \;\; Q_{\frac{3}{2}L}, \qquad \psi_{ij,T}^{(L)} = 0 \quad \text{on} \;\; \partial Q_{\frac{3}{2}L}.$$

6. For the indices

$$(i,j) \in \mathcal{J} = \{(1,2), (1,3), (2,3), (2,2), (3,3)\},$$

calculate

$$c_{ij,T}^{(L)} = -\int g \cdot \nabla\left(\sum_{k=1}^{3} \varphi_{k,T}^{(L)} \partial_k v_{h,ij}^{(L)} + (2 - \delta_{ij})\left(\psi_{ij,T}^{(L)} - \frac{a_{hij}^{(L)}}{a_{h11}^{(L)}}\psi_{11,T}^{(L)}\right)\right)$$

where $v_{h,ij}^{(L)}$ denote the $a_h^{(L)}$-harmonic polynomials

$$v_{h,ij}^{(L)} = \left(1 - \frac{1}{2}\delta_{ij}\right)\left(x_i x_j - \frac{a_{hij}^{(L)}}{a_{h11}^{(L)}} x_1^2\right).$$

7. Obtain $u_h^{(L)}$ as

$$u_h^{(L)} = \tilde{u}_h^{(L)} + \sum_{i=1}^{3}\left(\int g \cdot \nabla \varphi_{i,T}^{(L)}\right)\partial_i G_h^{(L)} + \sum_{(i,j)\in\mathcal{J}} c_{ij,T}^{(L)} \partial_{ij} G_h^{(L)}.$$

8. Solve for $u^{(L)}$ (here and for the rest of the paper we adopt Einstein's summation convention for repeated indices):

$$-\nabla \cdot a \nabla u^{(L)} = \nabla \cdot g \quad \text{in} \;\; Q_L, \qquad u^{(L)} = \left(1 + \varphi_{i,T}^{(L)}\partial_i + \psi_{ij,T}^{(L)}\partial_{ij}\right) u_h^{(L)} \quad \text{on} \;\; \partial Q_L.$$